

Dynamic MFA Activity: Decades in Candy Town

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Individual Handout: version 4, GNU General Public License v3.0

<https://github.com/treedmiller/candytown-mfa>

Activity Overview

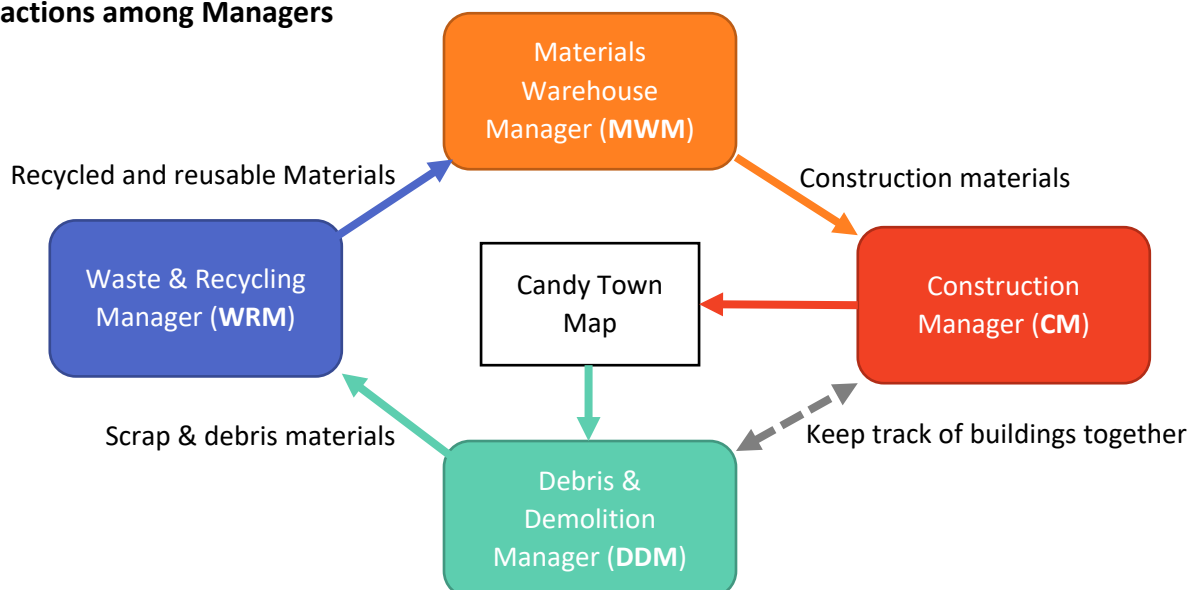
Candy Town started with just a few homes at the turn of the century, but it became a bustling town with lots of characters moving in and building houses and apartment buildings. Hummus became more popular than candy, though, so the population slowly dwindled, and decrepit old homes were torn down.

The purpose of this exercise is to practice a simplified bottom-up dynamic MFA to track the flows and stocks of residential buildings and associated materials over time. The PEZ candies will be used to represent building materials.

Roles

- **Materials Warehouse Manager (MWM)**
 - Give needed construction materials to **CM** to build homes (unwrap PEZ as needed)
 - Keep track of new and recycled materials available for construction
- **Construction Manager (CM)**
 - Get needed materials from **MWM** to build homes
 - Place materials on map to represent house being built
 - Keep track of buildings, working with **DDM**
- **Debris and Demolition Manager (DDM)**
 - Gather scrap from construction, deliver materials to **WRM**
 - Demolish buildings, deliver materials to **WRM**
 - Keep track of buildings, working with **CM**
- **Waste & Recycling Manager (WRM)**
 - Collect debris from construction from **DDM**
 - Deliver reusable or recyclable materials to **MWM**
 - Keep track of waste and recycling materials

Interactions among Managers



PEZ Materials Needed

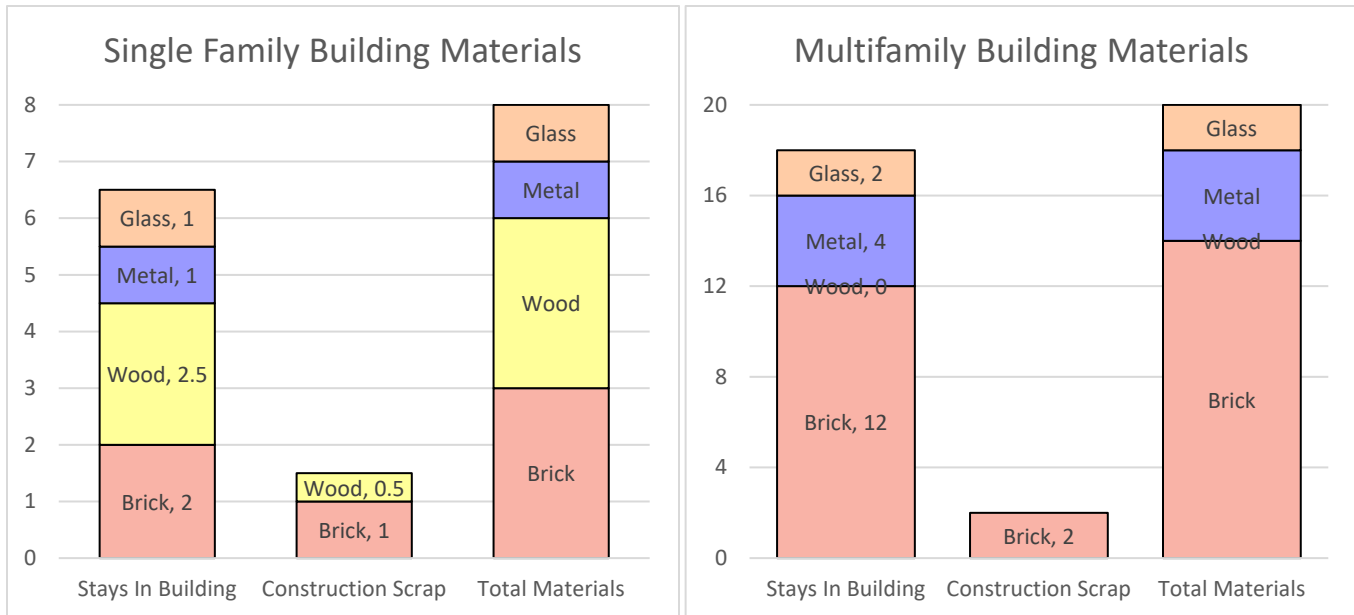
Each roll contains 12 PEZ.

Material	Brick	Wood	Metal	Glass
PEZ color	Red / Pink	Yellow	Purple	Orange
# PEZ rolls	7	2	2	2

Homes in Candy Town

There are two types of home in Candy Town, Single Family and Multifamily. The charts below show how many materials it takes to build each. The Single Family (SF) homes use wood, but the Multifamily (MF) homes do not.

While a SF home takes up one lot, each MF home spans two lots.



Additional Instructions

- Perform the various roles to follow the events in Candy Town as they unfold, moving and tracking materials.
- The **CM** must get enough materials from the **MWM** to account for the anticipated construction scrap. Calculate the Total Materials needed in the charts above.
- Scrap that is generated during construction is collected by the **DDM** and given to the **WRM**.
- Scrap brick and wood cannot be reused (assuming it is cut and no longer the standard shape suitable for future construction); therefore, all construction scrap goes into 'Combined Scrap & Debris for Landfill'.
 - (Assuming no metal or glass scrap during construction due to pre-fabrication.)
- After demolition, the **DDM** can send some materials to the **WRM** to be reused or recycled. The amount per house to be reused or recycled is shown in the table. The rest goes into 'Combined Scrap & Debris for Landfill'.
 - (Assuming that some of the bricks, wood, and metal are salvageable for reuse or recycling, but the rest of the demolished materials are either too damaged or impossible to separate from other materials.)

Per house	Brick	Wood	Metal	Glass
Single Family	1	2	1	0
Multi Family	6	0	2	0

- The reusable and recyclable materials collected by the **WRM** are sent to the **MWM** in the same time period, where they are mixed in with existing materials available for construction. Therefore, it will be recorded by the **WRM** as both a *Flow In* and a *Flow Out*, and a *Flow In* by the **MWM**.
 - (Assuming that metal has gone through recycling steps and is transformed back into its standard shape.)

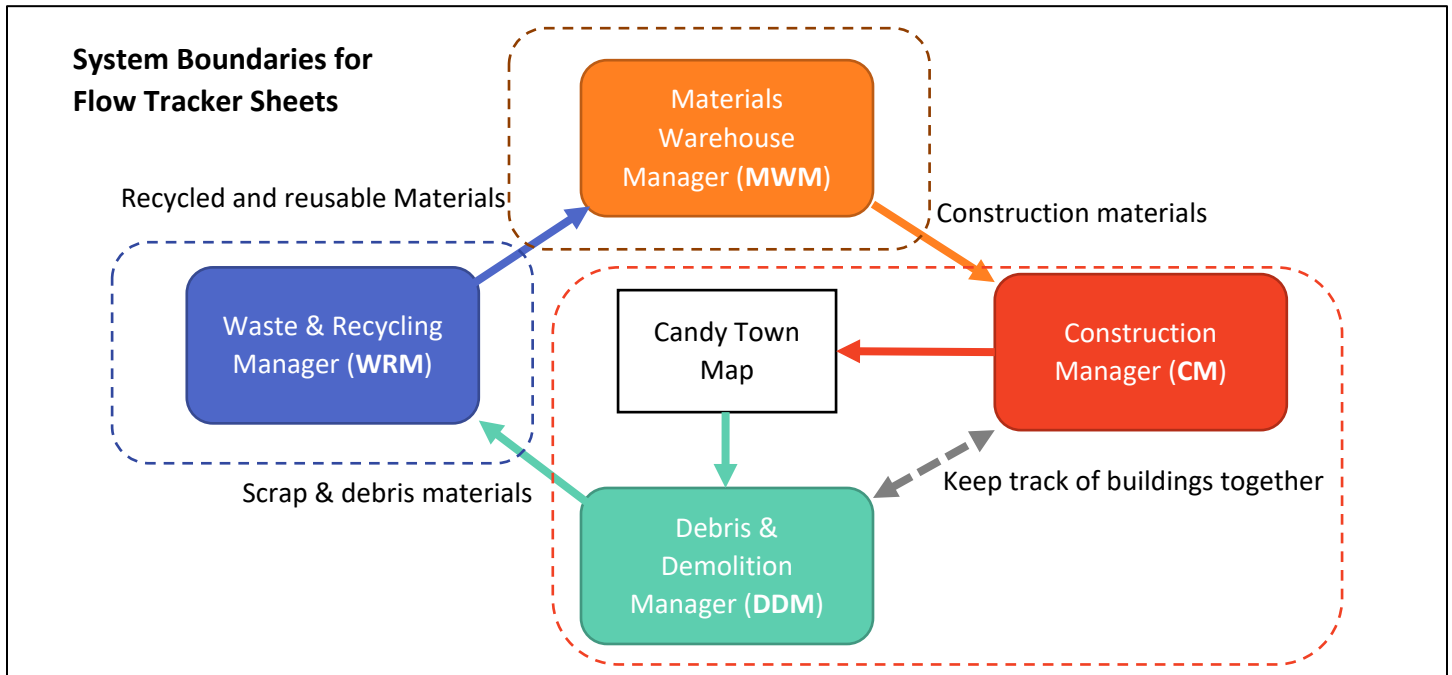
Events in Candy Town

- 2000
 - The first SF home in Candy Town is built at 1 Sweet Street.
- 2010
 - A new neighbor built another SF home at 2 Sweet Street.
- 2020
 - Their friends moved to town and built SF homes at 3 and 4 Sweet Street.
- 2030
 - As word started to spread, two more SF homes were built at 5 and 6 Sweet Street.
 - With popularity growing, the first MF buildings were built to accommodate more people. The builder put two of them near each other, at 1-3 and 5-7 Sugar Ave.
- 2040
 - The home at 2 Sweet Street was tragically struck by lightning and partially burned (no one was hurt!). The home next door caught fire, too. They were both demolished, but some of the materials were still structurally sound and could be recovered.
 - Another builder decided to build two more MF homes across the street from the existing ones, at 2-4 and 6-8 Sugar Ave.
- 2050
 - For the Golden Jubilee, the 50th Anniversary of Candy Town, they took stock of all the materials in their buildings. Count how many materials are present across all of Candy Town's buildings. Take note on the Flow Tracker for CM & DDM in the specified row.



- 2060
 - The Candy Town historic preservation funds ran out. Although they tried to keep the remaining SF homes in Candy Town in good shape, they all fell into disrepair and needed to be demolished.
- 2070
 - Nothing is built or demolished.
Everyone in Candy Town now lived in MF homes. (They are very energy efficient!)
- 2080
 - The economy started declining as healthy snacks became more popular than candy, and neighbors all left Candy Town and headed to Hummus Town. The remaining MF homes were abandoned, and demolished.
 - The usable materials are carted off to the recently established Hummus Town, but the landfill stays.

Calculations and Reflections



Calculations

- Calculate the Stock Change and Stock on the Flow Tracking Sheets, according to the System Boundaries above
- Compare to the calculated stock to the stock count on the 50th Anniversary Golden Jubilee.
- Complete the Lifespan Distribution Handout
- Go to spreadsheet and fill in the boxes using the flow tracker sheets

Reflection Questions (please discuss as pairs or a group and jot some notes)

1. Now that the Industrial Ecologists in Hummus Town have an estimated lifespan distribution based on Candy Town buildings, how might they use it? What insights could they gain, and how might it help with planning?
2. Not all the demolished materials were reused or recycled.
 - a. If reuse and recycling rates were increased in Candy Town, could reused and recycled materials have met the demand for materials over time?
 - b. Are there implications for how we think of reuse and recycling in the 'real world'?
 - c. Does this thinking apply to all materials, or only some kinds?
3. How could Candy Town take advantage of the stock and flow data recorded?
4. Each of the managers recorded flow categories differently.
 - a. Which manager's data should align with another's? Test this your theory for a couple years.
 - b. In real life, different stakeholders often record things with different levels of aggregation and need to be carefully pieced together for a comprehensive MFA. Why might that be, and what could be done?